

Eindhoven University of Technology

AI-Supported Sustainability Self-Assessment for TU/e Research

ENSU – Written Assignment Stage 1

Course: 5ARF0 Ethics of AI: Theory and Practice
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Main-text count excludes headings, tables, figures, references, appendices and draft placeholders.

1 Introduction and Motivation

1.1 Topic, Motivation, Values and Needs

According to TU/e's Sustainability Visioning Report, TU/e has already seen sustainable development as a major part of campus development and trying to achieve the goal of fully embodying sustainability and proactively engaging with societal challenges in 2050. However, the researchers need to put these into practical activities; this refers to specific equipment and data, which is often confusing. This project is trying to explore whether AI can help in this progress while remaining transparent and supporting human judgement.

1.2 Objectives and Key Terms

In this project, AI support tool is proposed to help users judge and examine the potential environmental impacts, taking equipment use, runtime, electricity consumption and other relevant factors into account.

A large language model (LLM) is a language model with a large number of parameters, trained on extensive text data to support language understanding and generation. Here, LLM would provide supported suggestions based on the provided description of activities and use the formulas to calculate the environmental impacts through a separate module.

The design objective is trying to help the researchers get reliable self-reflection on their own project. The proposed workflow comprises plan submission, AI-assisted activity analysis, formula-based impact estimation, comparison with relevant reference cases, and feedback with practical recommendations. Any assessment score would require agreed criteria. The database will be based on the known data and results of the papers, and will also include some examples, aiming to be professional and reliable. Missing information should trigger questions or qualified feedback. Evaluation would examine evidence traceability, consistency and whether users find the feedback helpful when revising their plans.

1.3 Context and Scope

The project focuses on the Research domain at TU/e. Its intended users are researchers, including students conducting research, who want to reflect on the environmental impacts of their planned research activities. This tool is designed for the planning and revising the research activities, such as data processing and computational experiments. The assessment covers selected environmental impacts for which sufficient activity data and reference information are available. The tool will not be used to manage campus facilities, evaluate university curricula, or decide which projects receive approval or funding. It will not provide a complete life-cycle assessment or rank researchers and projects across disciplines.

1.4 Tools, Methods and Frameworks

[TODO: Explain why Design for Values or Value Sensitive Design fits the aim. Introduce conceptual, empirical, and technical investigations with references.]

Logic: research context → stakeholder need → value → requirement → tool feature → success indicator.

2 Initial Results

2.1 Preliminary Literature Search

[TODO: Report databases, search terms, dates, selection criteria, number of sources, and main evidence gaps.]

Source	Type	Main finding	Need or value	Design implication
[TODO: Citation]	[TODO: Type]	[TODO: Finding]	[TODO: Need/value]	[TODO: Implication]
[TODO: Citation]	[TODO: Type]	[TODO: Finding]	[TODO: Need/value]	[TODO: Implication]

Table 1: Literature-search log.

Preliminary findings. [TODO: Synthesize researcher needs, current assessment methods, AI opportunities, known risks, and evidence gaps.]

2.2 Design Strategy

[TODO: Describe the steps from evidence and stakeholder input to requirements, mock-up, expert review, evaluation, and revision.]

2.3 Initial Requirements and Constraints

ID	Type	Requirement or constraint	Rationale or source	How to check it
R1	Functional	The tool needs to allow researchers to describe their work in their own words.	This is important to make the tool easily usable.	Test with example research projects.
R2	Functional	The tool has to identify the impact of the research on sustainability.	This is the main purpose of the tool.	Provide the tool with examples of research that have different sustainability impacts.
R3	Constraint	The tool needs to make clear that it is not an official approval, but a guideline.	Sustainability is complex and AI alone will not be enough to make a final judgement.	Check whether the tool clearly communicates its advisory nature and requires human review of the output.
R4	Constraint	The researcher is responsible for changes made according to the advice from the tool.	The users are performing the research and therefore remain responsible for their decisions.	Check whether responsibility for decisions remains with the researcher.

Table 2: Initial requirements and constraints

Identifying the sustainability impacts is the most important requirement mentioned above. This is what the initial goal was for the tool. Researchers, including students performing research, will

use the tool to find out what parts of their research potentially form a problem for sustainability. Environmental, social and economic considerations inform the sustainability discussion; the initial assessment is limited to selected environmental impacts, as specified in Section 1.3. The output of the tool is only a guideline for the user so that they can improve their research. The responsibility therefore stays with the researcher.

2.4 Anticipated Function and Purpose of the Tool

- **User and immediate goal:** TU/e researchers wanting to know the sustainability impact of their research.
- **Input:** A precise description of the research.
- **System actions:** The tool analyzes the research for potential sustainability impacts and, if needed, asks follow-up questions.
- **Output:** A guideline for researchers explaining how their research impacts sustainability.
- **Human review or correction:** Researchers need to carefully review the feedback from the tool.
- **Handling of missing information or uncertainty:** Missing information prompts follow-up questions; unresolved gaps are reported and unsupported estimates are withheld.

For this scenario, a Tu/e researcher describes their work to the tool. Their research is about developing a new battery for electric cars. Potential impacts can be, for example, the gathering and production of required materials for the battery. The production will of course require energy and potentially lots of costs. The tool will also look at the environmental impacts of the production. The tool will ask questions about the materials used for the production of the new battery. With this information, the tool can then ask further questions about the costs, energy, etc. When the tool has all the necessary information, it can then provide a detailed guideline for the research containing the potential impact with explanations and possible improvements. It is now up to the researchers to review the guidelines from the tool and decide what they want to change. The AI is not responsible for these decisions, as it makes clear that the output is not an official approval for the research.

Figure 1 summarises the proposed workflow; Figure 2 applies it to the battery research example. AI would organise descriptions and draft source-supported suggestions, while explicit calculations would be performed by a separate module. Unresolved data gaps would be disclosed and unsupported estimates withheld.

A proposed TU/e sustainability archive would store comparable research cases with activity conditions, model and advice versions, and follow-up evidence under pseudonymous identifiers. For battery research, records could link a change in a fabrication procedure to subsequent material use and electricity measurements. AI could assist with organising records and identifying relevant cases; an authorised campus curator would approve evidence and guidance updates. Researcher-confirmed outcomes would inform later suggestions, subject to consent and comparability checks. Score changes would be recorded only under agreed criteria and would not, by themselves, demonstrate environmental improvement. Archive roles and resources remain to be agreed with the client.

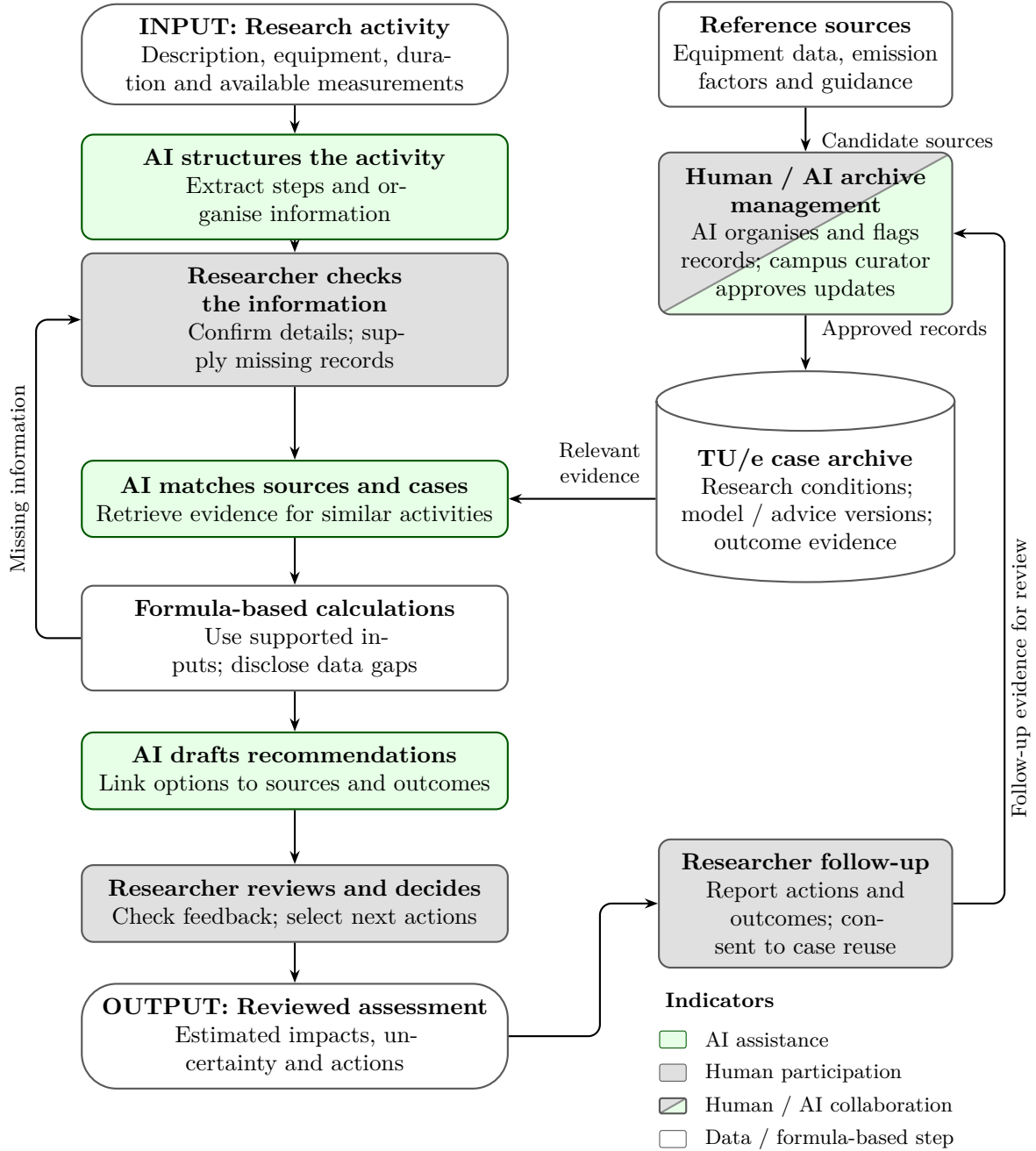


Figure 1: Proposed research self-assessment workflow with human-reviewed reuse of campus case evidence.

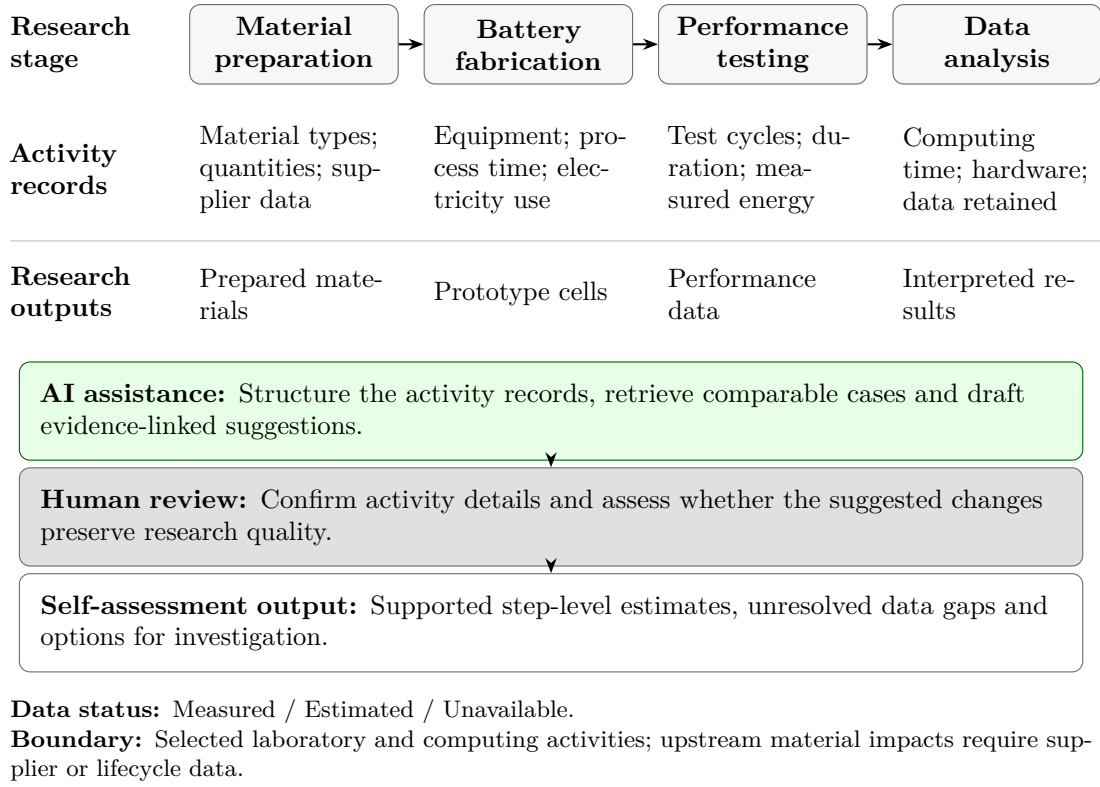


Figure 2: Illustrative battery research stages, records and outputs used for self-assessment. Green denotes AI assistance; grey denotes human review.

2.5 Projected Users, Stakeholders and Values

Stakeholder	Role or relationship to the tool	Expected benefit	Concern or burden
TU/e researchers	Main users	Identify impact on sustainability	Takes time providing all the information
Students	Main users	Same as researchers	Maybe depending too much on the tool
Supervisors	Support	Get a better understanding to help	Needs to spend extra time reviewing the output of the tool
TU/e sustainability team	Experts	Validate the tool	Not always available

Table 3: Stakeholder notes

The Tu/e researchers and students will be the main users of our tool. They provide the tool with their research and receive sustainability guidance for their project in return. Supervisors for the projects performed by the researchers and students will also use this information from the tool but will not need to use the tool directly. Some stakeholders might have different expectations from the tool, but these are the benefits it is intended to provide.

3 Discussion

3.1 Ethical Issues and Value Conflicts

The definition of sustainable research determines how activities are assessed. Hasselbalch and van Wynsberghe [2025] locate sustainability within sociotechnical relations of power, making the selection of criteria an ethical issue. If electricity consumption is prioritised, an intensive experiment may be rated unfavourably despite its scientific value. In the proposed assessment, environmental indicators would therefore be presented alongside explanations of research purpose. Comparisons would be restricted to activities with comparable objectives and quality requirements. Client agreement is needed on the weighting of criteria and the appropriateness of comparisons across disciplines. A single aggregate score could otherwise conceal disagreements about research priorities.

Numerical estimates may be treated as more authoritative than their evidence warrants. Lacoste et al. [2019] identify hardware, runtime and electricity supply as factors affecting emissions from machine learning training. Their method does not establish validity for every research activity. Apparent differences between cases may be smaller than calculation uncertainty. Measured and estimated values would therefore be distinguished, with missing information flagged and sources and boundaries disclosed. Input correction would remain available to researchers. These provisions support scrutiny, but cannot ensure that an assessment is critically examined.

The resource consumption of the assessment tool also requires justification. Lin [2023] argues for transparency about AI's energy use. Explicit calculations would be assigned to fixed formulas, while AI assistance would support descriptions and suggestions, including retrieval from the reviewed archive. Comparison with a structured non-AI assessment is needed to establish whether any additional benefit warrants the computation and review effort. Environmental improvement has not yet been demonstrated.

3.2 Design Barriers and Challenges

Defining a manageable assessment boundary is a practical constraint. Consumption from shared equipment and computing services may be difficult to attribute to individual activities. One representative workflow would be mapped with the client to establish which records are available. Estimates of equipment operation would state their exclusions. Replacement advice would require evidence about manufacturing impacts as well as operating savings.

Reference data and recommendations require maintenance. Sources, review dates and version histories would be recorded so that earlier assessments remain traceable. Following the institutional perspective of Hasselbalch and van Wynsberghe [2025], responsibility for correcting or withdrawing outdated guidance would need to be assigned at TU/e. The responsible role and available resources remain unresolved. For the proposed campus archive, outcome evidence and comparable activity conditions would require review before cases were reused. Higher assessment scores alone would not demonstrate reduced environmental impact. Reported outcomes would be distinguished from verified measurements, since selective reporting could favour successful cases and bias recommendations for future users of the proposed archive.

Evaluation would compare AI-assisted feedback with a structured checklist on the same cases. Evidence support, omitted issues and correction effort would be examined to establish whether further development is justified.

Risk	Stakeholders	Values	Response and basis	Limitation
Privacy	Researchers; case contributors	Privacy; autonomy	Minimise stored details; control archive access and case reuse [Hasselbalch and van Wynsberghe, 2025].	Activity descriptions may identify a contributor.
Unclear AI advice	Researchers	Transparency; accountability	Show sources, model version and uncertainty [Lacoste et al., 2019].	A retrieved case may be irrelevant or misinterpreted.
Greenwashing	TU/e; researchers	Sustainability; trust	State boundaries; check observed impacts alongside score changes [Lin, 2023].	Better scores need not reflect lower overall impacts.
Rebound effects	TU/e; affected communities	Environmental impact	Track total activity and resource use alongside efficiency [Luccioni et al., 2025].	Savings may encourage additional activity.
Unfair recom- mendations	Research groups	Fairness	Compare similar objectives and conditions; review archive coverage [Hasselbalch and van Wynsberghe, 2025].	Uneven case contributions can distort comparisons.
Reduced human control	Researchers; campus curator	Human oversight	Keep decisions with researchers; assign responsibility for archive updates [Santoni de Sio and van den Hoven, 2018].	Formal review does not ensure critical engagement.

Table 4: Anticipated ethical risks, literature-informed responses and remaining limitations.

References

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- Filippo Santoni de Sio and Jeroen van den Hoven. Meaningful human control over autonomous systems: A philosophical account. *Frontiers in Robotics and AI*, 5:15, 2018. doi: 10.3389/frobt.2018.00015.

A Documentation of Process

[TODO: Record meetings, research, design work, writing work, decisions, and evidence.]

Date	Participants	Activity	Output or decision	Next action
18 Sep 2026	Yehao Chen; AI writing assistance	Revise discussion; diagrams	Dis-draft Proposed workflow; revised text	Group review of scope and wording
[TODO: Date]	[TODO: Names]	[TODO: Activity]	[TODO: Output]	[TODO: Action]

Table 5: Project process log.

B Project Planning

Date	Activity	Owner	Deliverable
18 Sep, 23:59	Integrate and submit Phase 1	[TODO: Owner]	Phase 1 report
22–29 Sep	Confirm scope and evidence needs during coaching	[TODO: Owner]	Revised requirements
30 Sep–10 Oct	Develop mock-up; prepare evaluation and practice pitch	[TODO: Owner]	Mock-up and evaluation plan
11 Oct, 23:59	Submit Phase 2 draft for peer review	[TODO: Owner]	Draft report
16 Oct, 23:59	Complete individual peer review	Each member	Peer feedback
17–20 Oct	Revise and record pitch	[TODO: Owner]	Video pitch: 20 Oct, 23:59
23 Oct, 23:59	Submit Phase 2 final version	[TODO: Owner]	Final report
24 Oct, 23:59	Complete groupmate evaluation	Each member	Individual evaluation

Table 6: Proposed remaining project schedule (2026).